

ETHERNET STANDARD AND SPECIFICATION

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Pins:	1	2	3	4	5	6	7	8
T568A:	W/G	G	W/O	BL	W/BL	O	W/BR	BR
T568B:	W/O	O	W/G	BL	W/BL	G	W/BR	BR

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	CAT3, CAT5, CAT5e	100 meters	RJ-45
100BASE-TX (Fast Ethernet)	100 Mbps	CAT5 or CAT5e	100 meters	RJ-45
1000BASE-T (Gigabit Ethernet)	1000 Mbps (1 Gbps)	CAT5e, CAT6, CAT6A	100 meters	RJ-45

**3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type..**

To connect two laptops directly using an Ethernet cable, I would use a crossover cable.

Reason:

A crossover cable swaps the transmit (Tx) and receive (Rx) wire pairs, allowing the two laptops to communicate directly without a switch or router. This is the traditional method for connecting two similar devices.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Device	Location	Likely Ethernet Standard
Desktop Computer	Computer Lab / Home	1000BASE-T (1 Gbps)
Wi-Fi Router	Home	1000BASE-T (1 Gbps)
Network Printer	College Office / Computer Lab	100BASE-TX (100 Mbps)